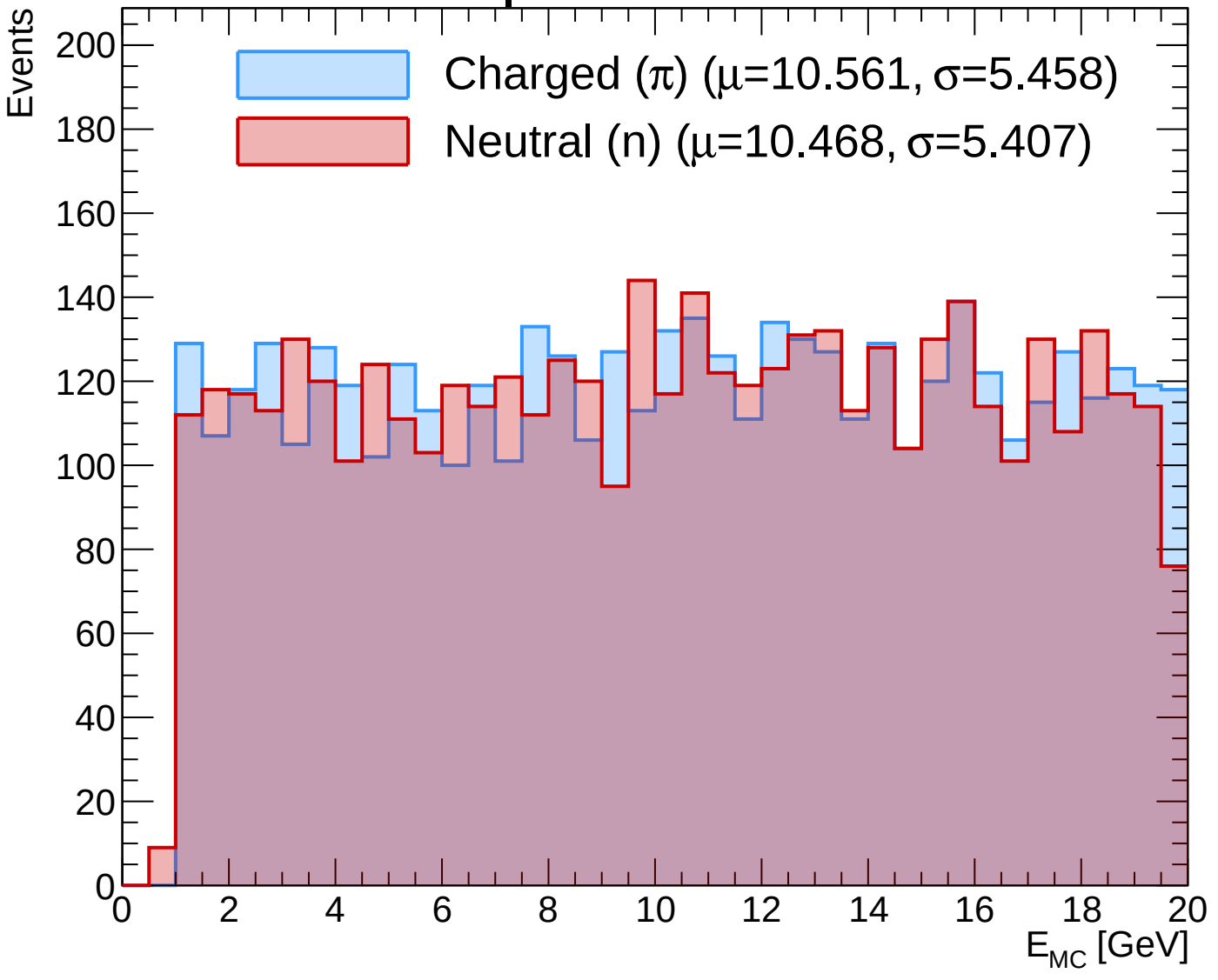
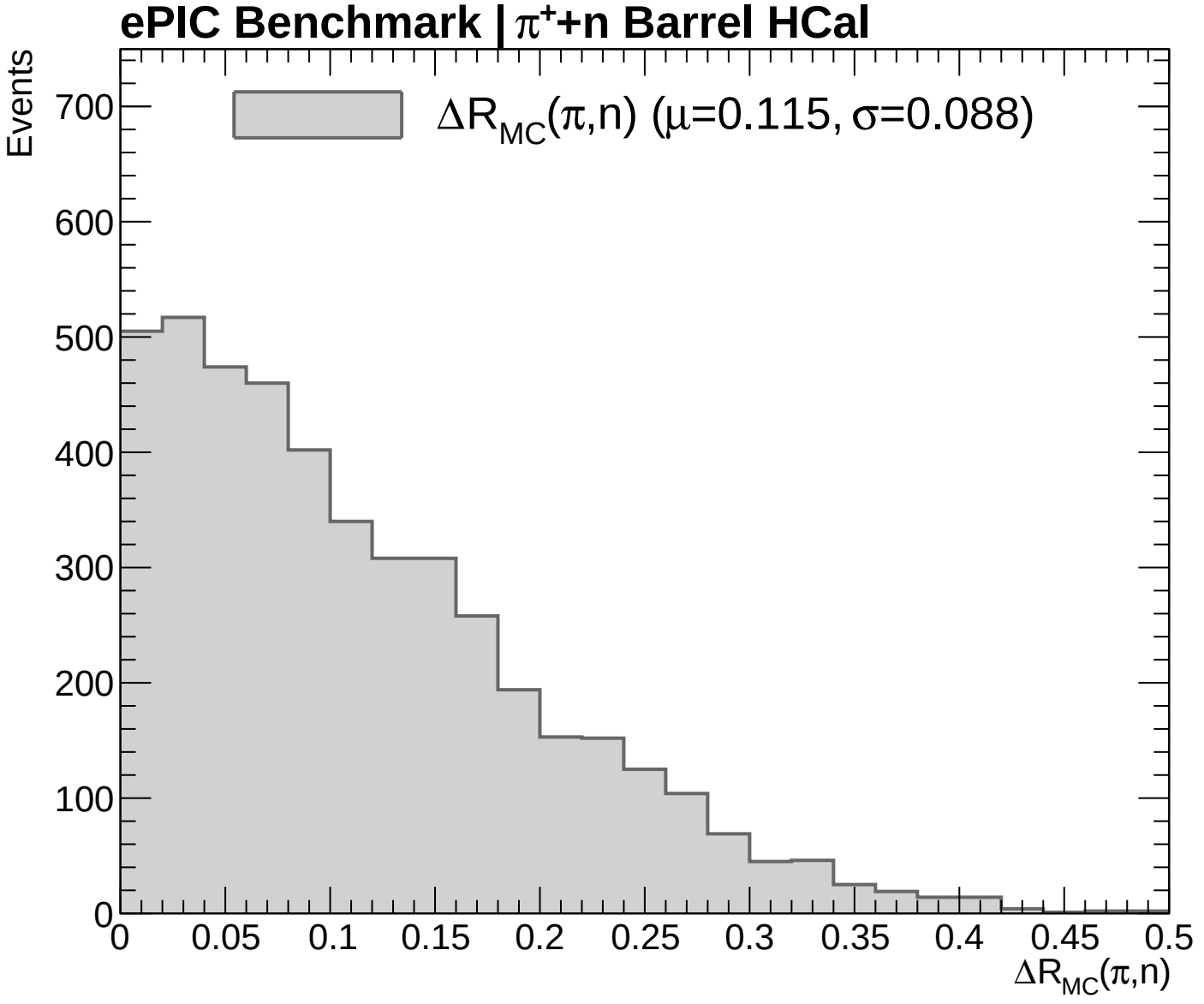
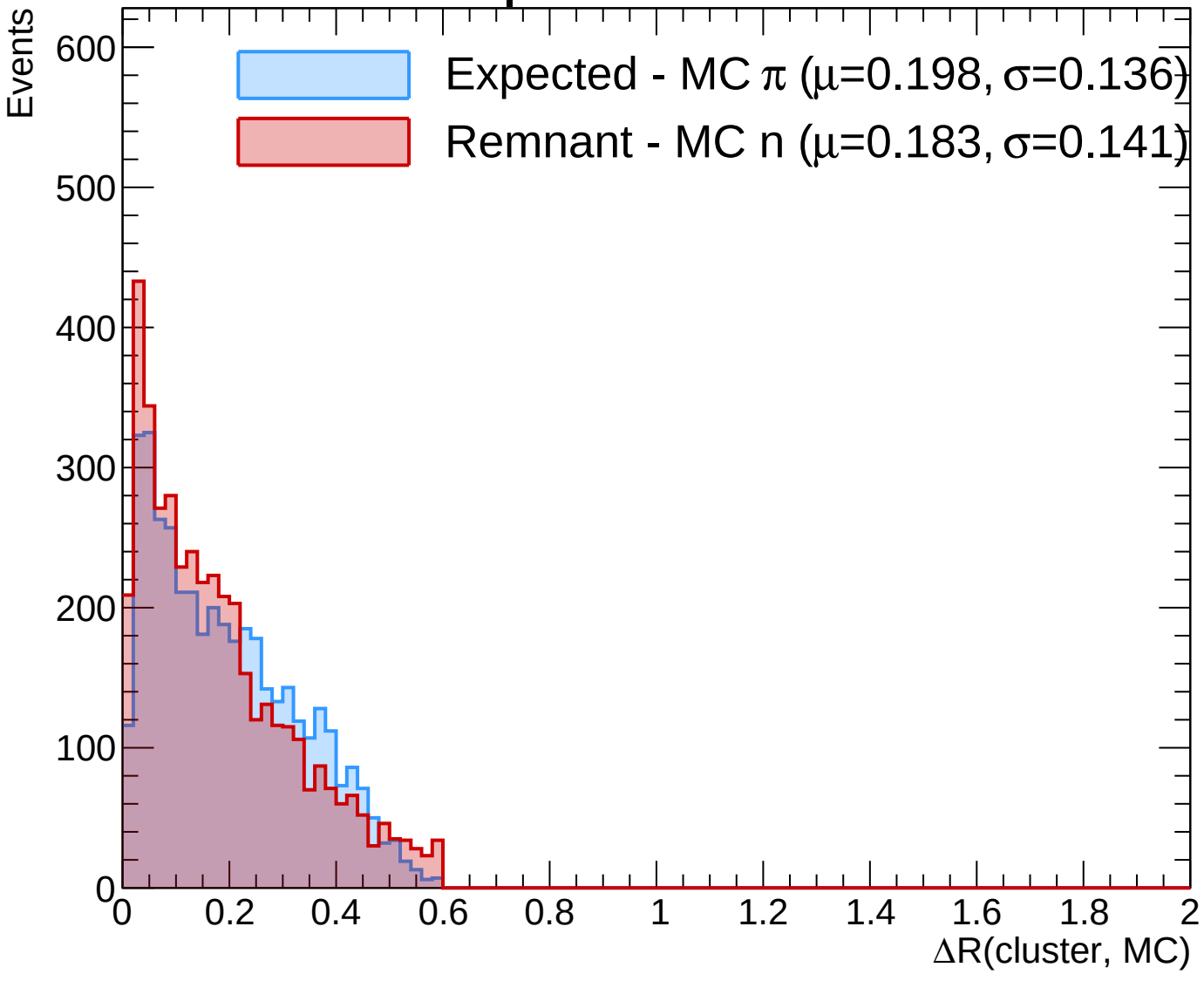
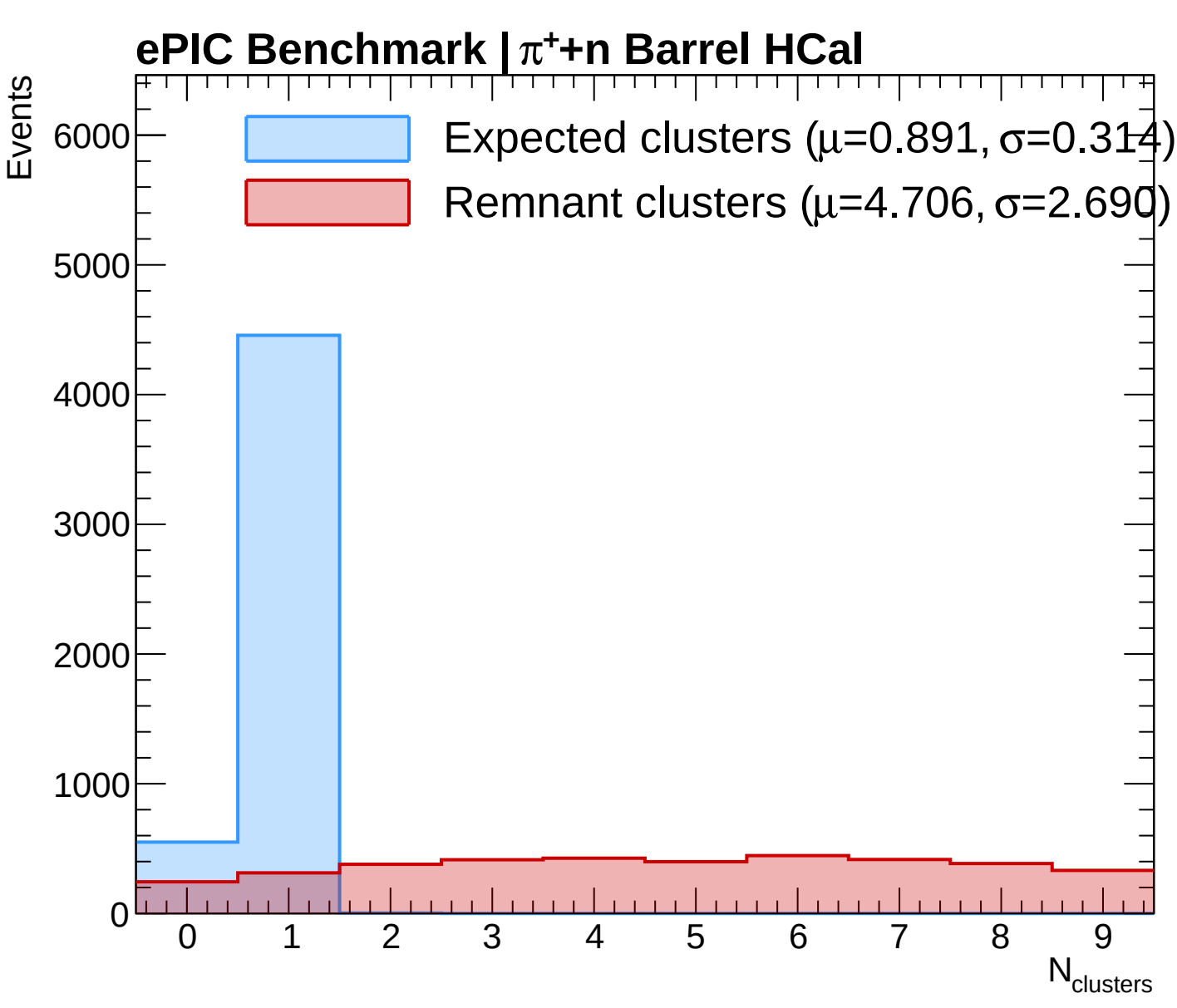
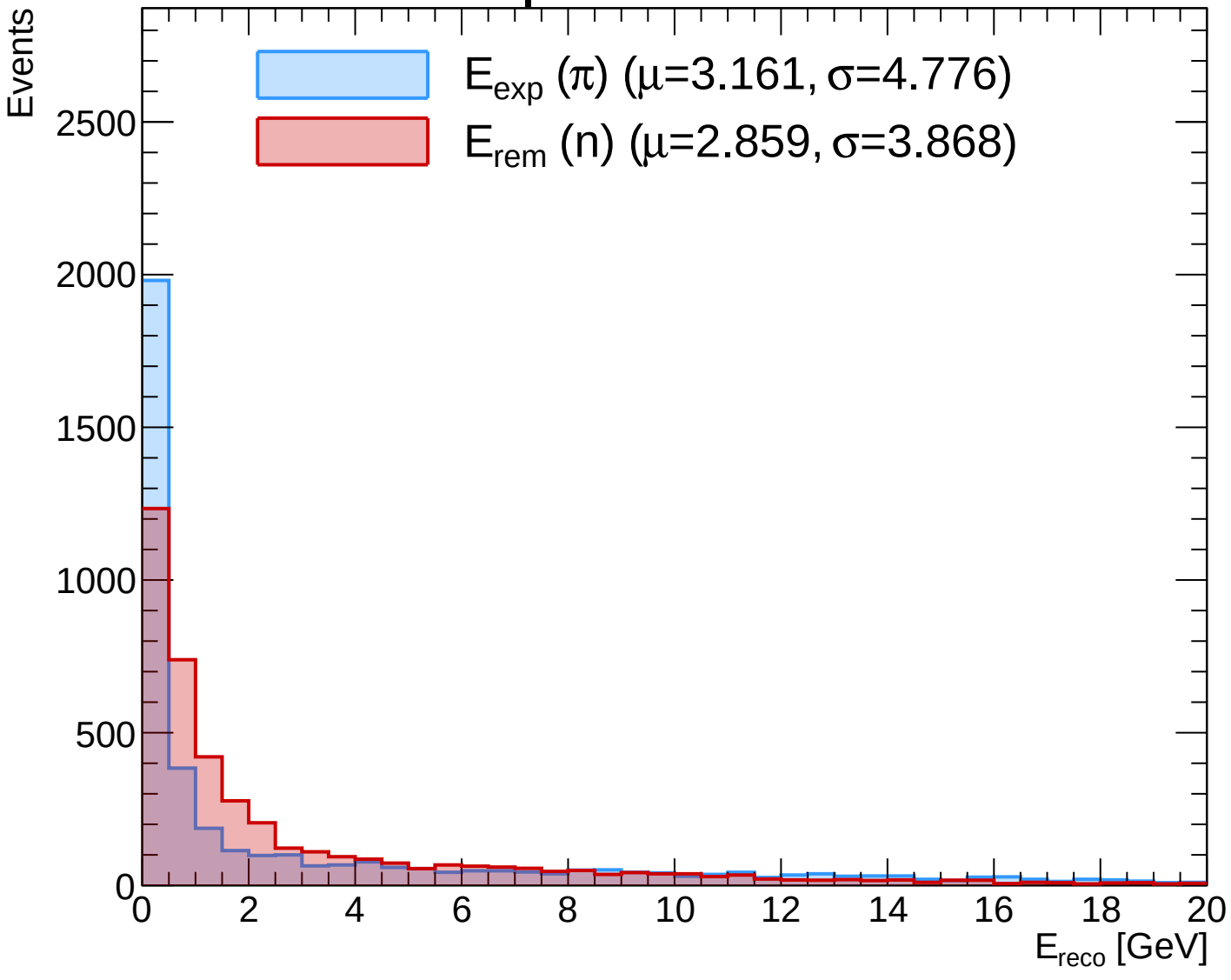
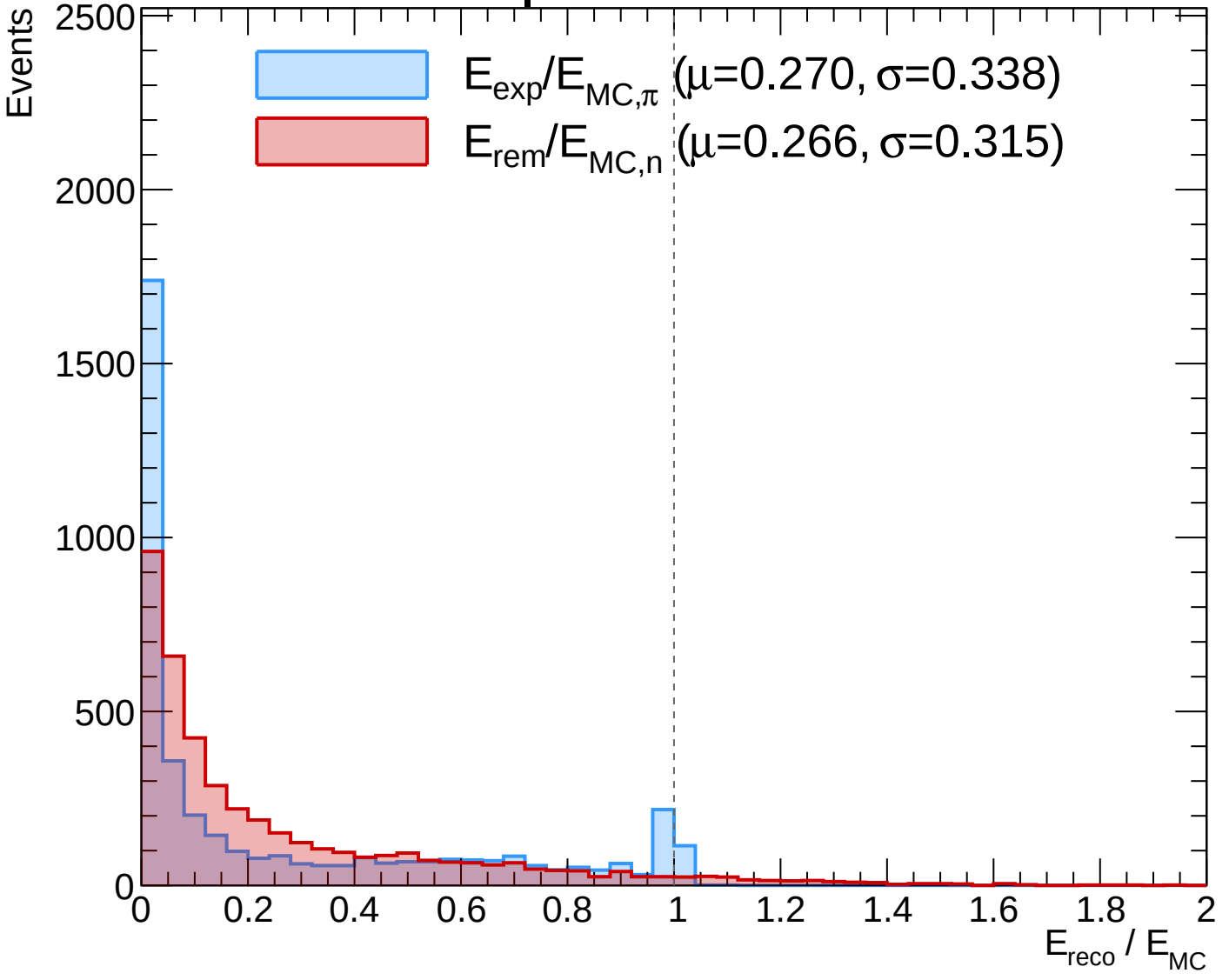

ePIC Benchmark | π⁺+n Barrel HCal
Charged (π) (μ=10.561, σ=5.458)
Neutral (n) (μ=10.468, σ=5.407)
Events
E_MC [GeV]
0
20
40
60
80
100
120
140
160
180
200
0
2
4
6
8
10
12
14
16
18
20

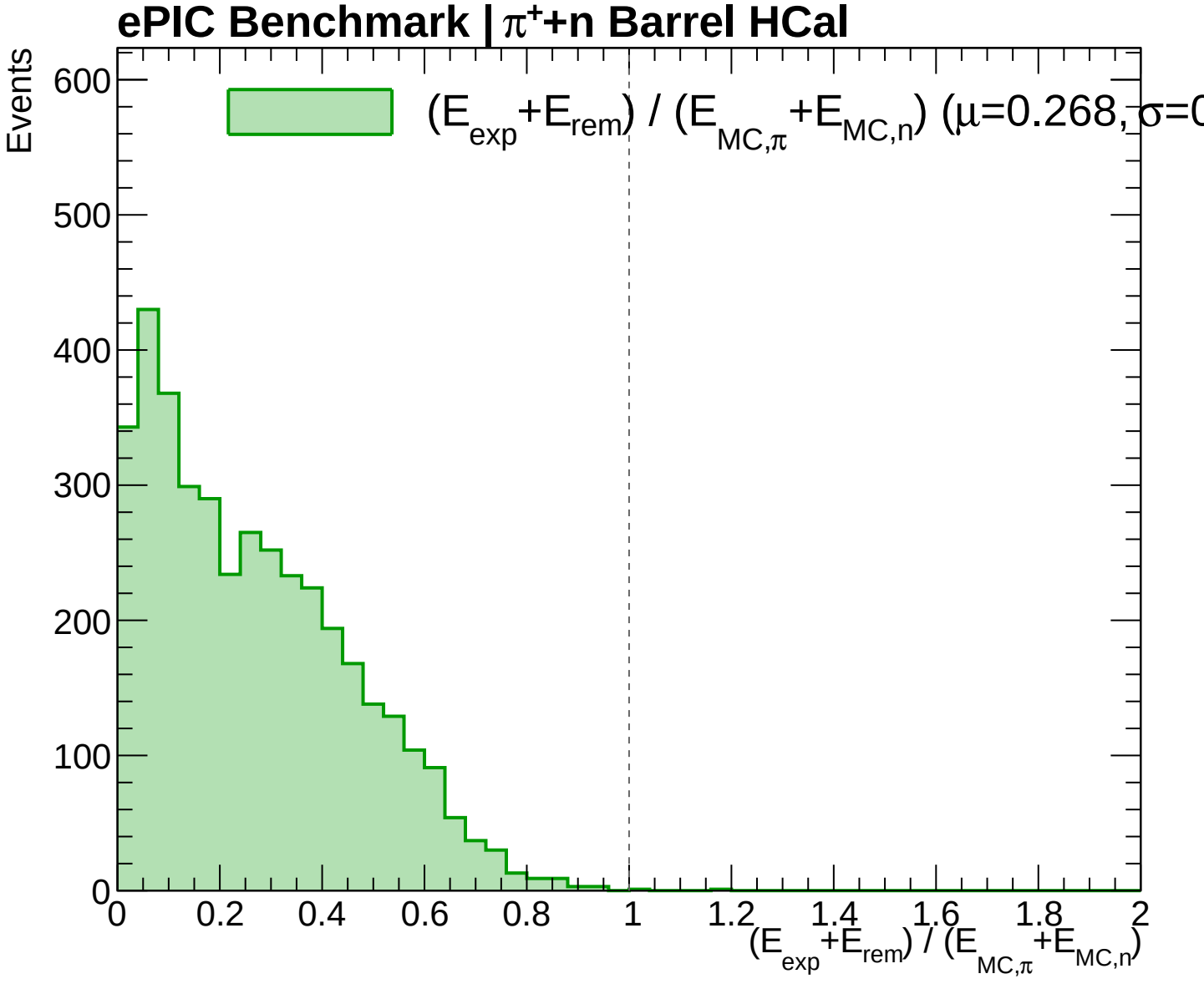

ePIC Benchmark | π⁺+n Barrel HCal
Events
ΔR_MC(π,n) (μ=0.115, σ=0.088)
700
600
500
400
300
200
100
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(π,n)

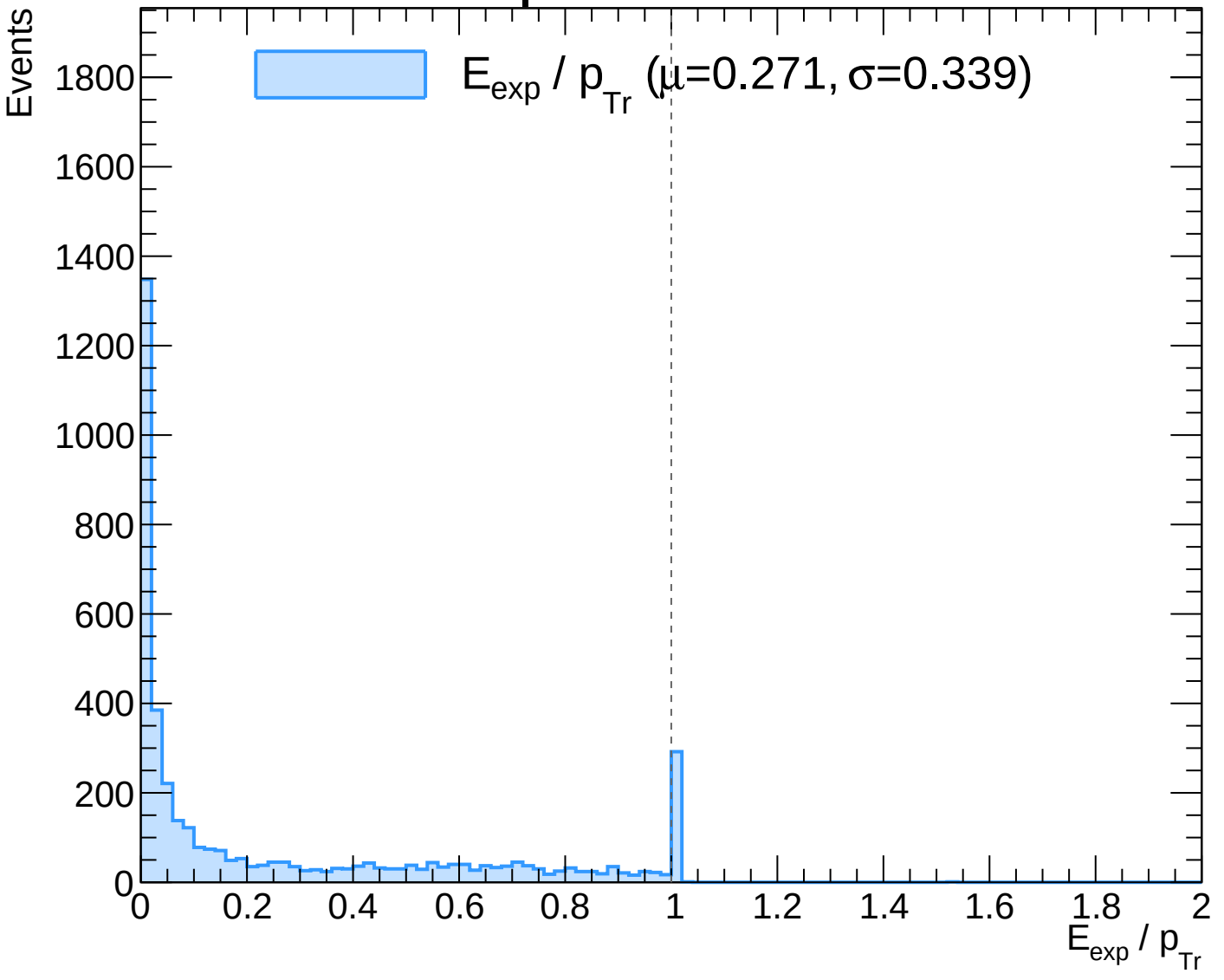

ePIC Benchmark | π⁺+n Barrel HCal
Events
Expected - MC π (μ=0.198, σ=0.136)
Remnant - MC n (μ=0.183, σ=0.141)
600
500
400
300
200
100
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
ΔR(cluster, MC)

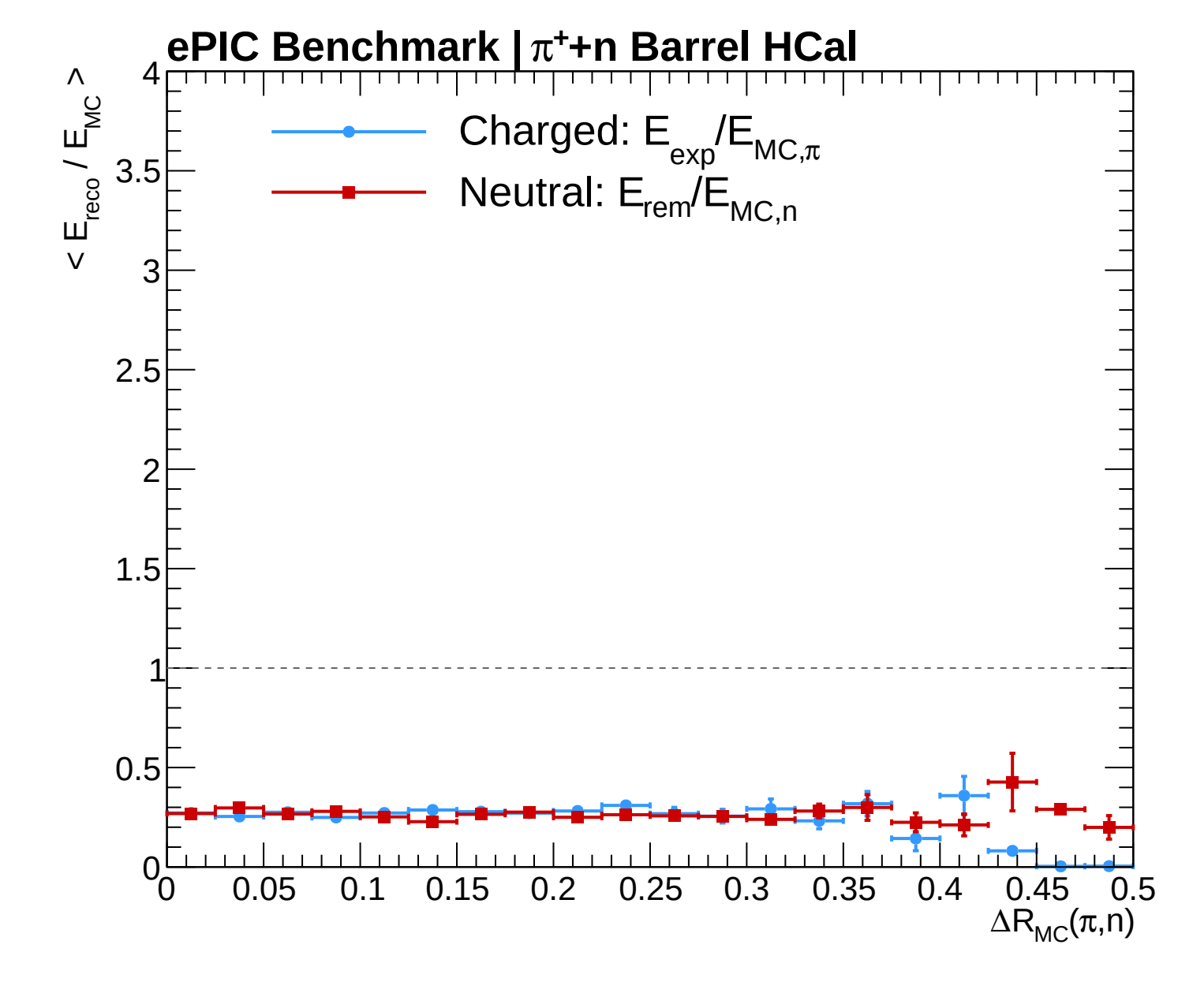

ePIC Benchmark | π⁺+n Barrel HCal
Events
6000
5000
4000
3000
2000
1000
0
0
1
2
3
4
5
6
7
8
9
N_clusters
Expected clusters (μ=0.891, σ=0.314)
Remnant clusters (μ=4.706, σ=2.690)

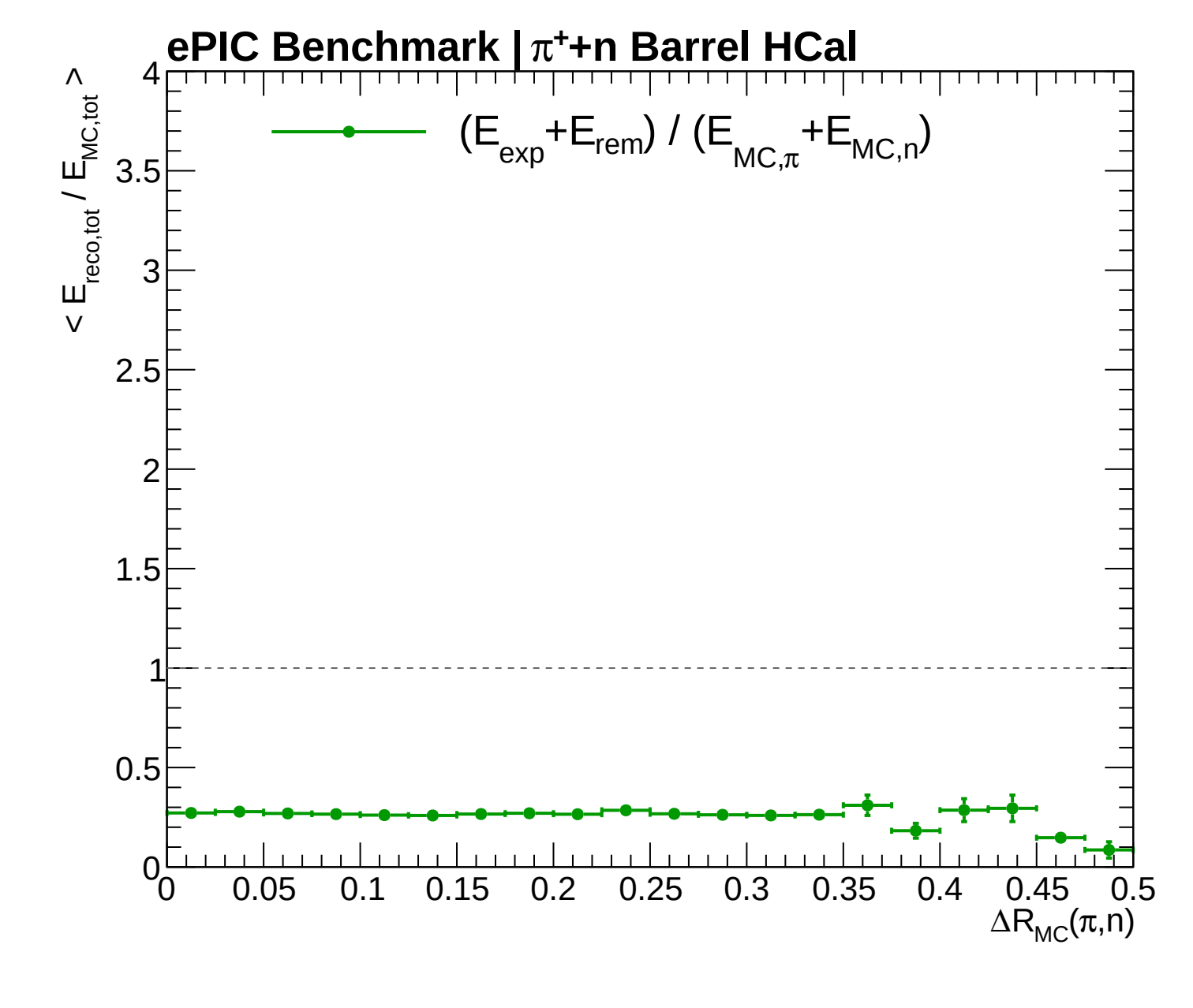

ePIC Benchmark | π⁺+n Barrel HCal
Events
E_exp (π) (μ=3.161, σ=4.776)
E_rem (n) (μ=2.859, σ=3.868)
2500
2000
1500
1000
500
0
0
2
4
6
8
10
12
14
16
18
20
E_reco [GeV]

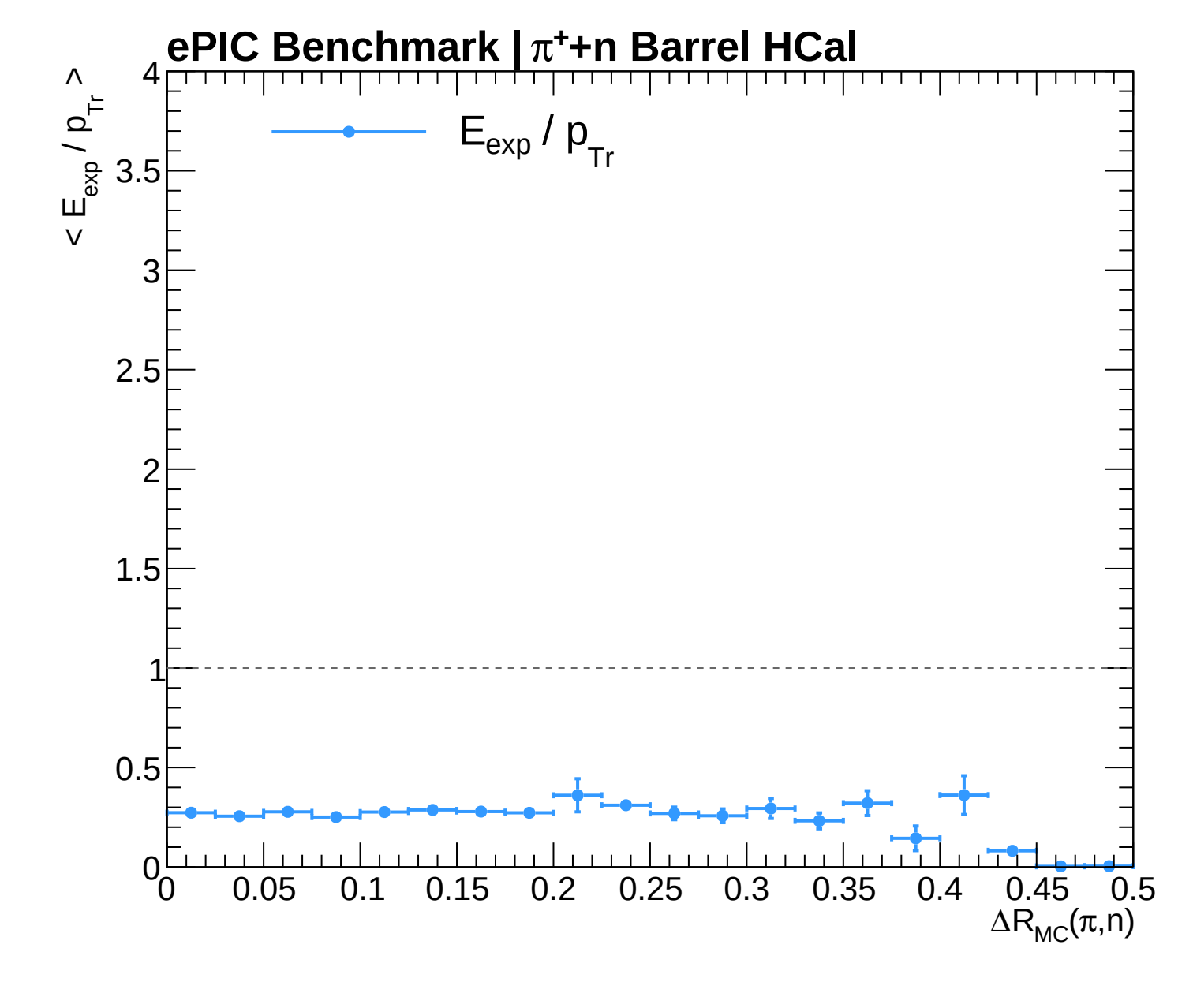

ePIC Benchmark | π⁺+n Barrel HCal
Events
2500
2000
1500
1000
500
0
E_exp/E_MC,π (μ=0.270, σ=0.338)
E_rem/E_MC,n (μ=0.266, σ=0.315)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_reco / E_MC

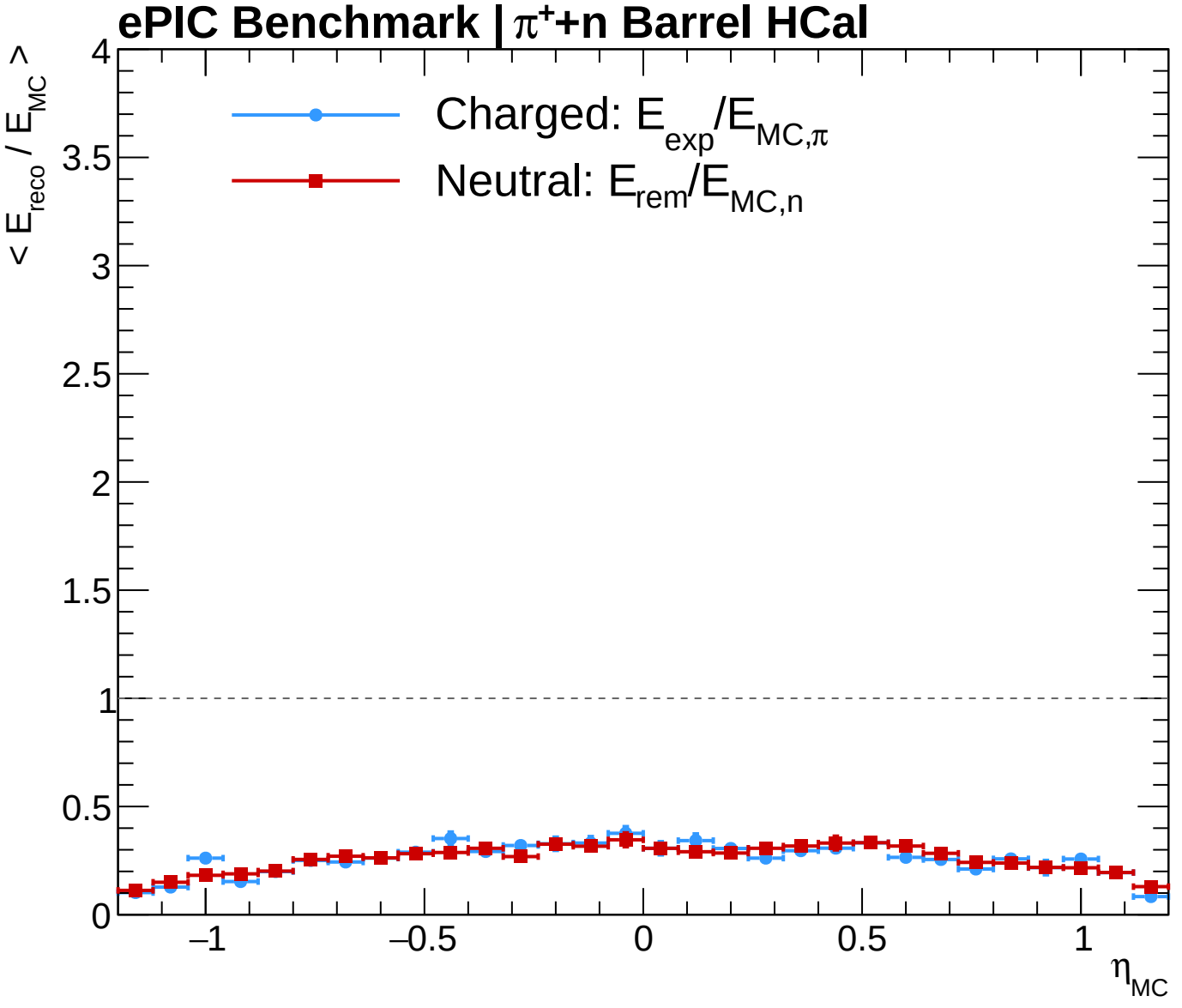

ePIC Benchmark | π⁺+n Barrel HCal
Events
600
500
400
300
200
100
0
(E_exp+E_rem) / (E_MC,π+E_MC,n) (μ=0.268, σ=0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
(E_exp+E_rem) / (E_MC,π+E_MC,n)

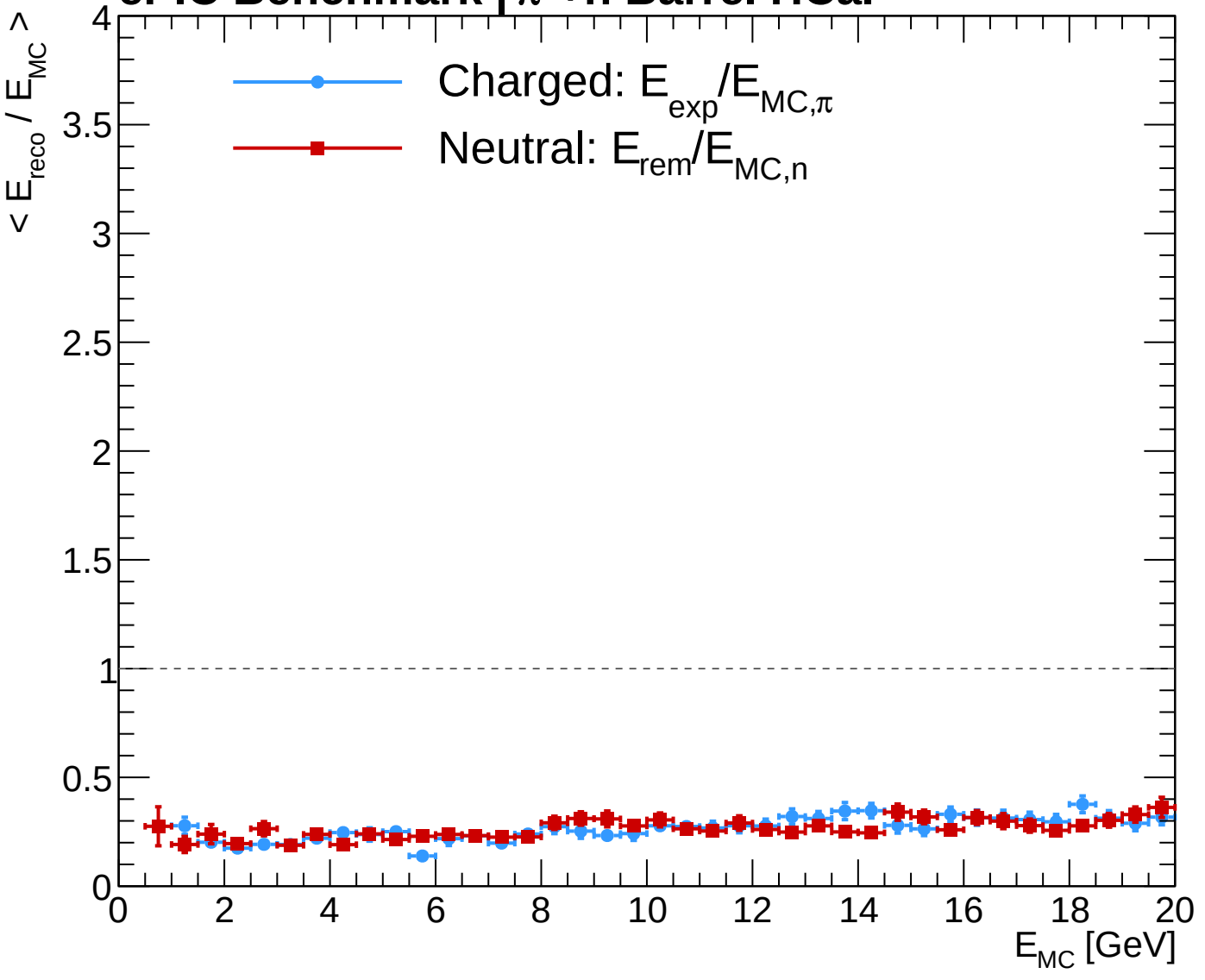

ePIC Benchmark | π⁺+n Barrel HCal
Events
E_exp / p_Tr (μ=0.271, σ=0.339)
1800
1600
1400
1200
1000
800
600
400
200
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_exp / p_Tr

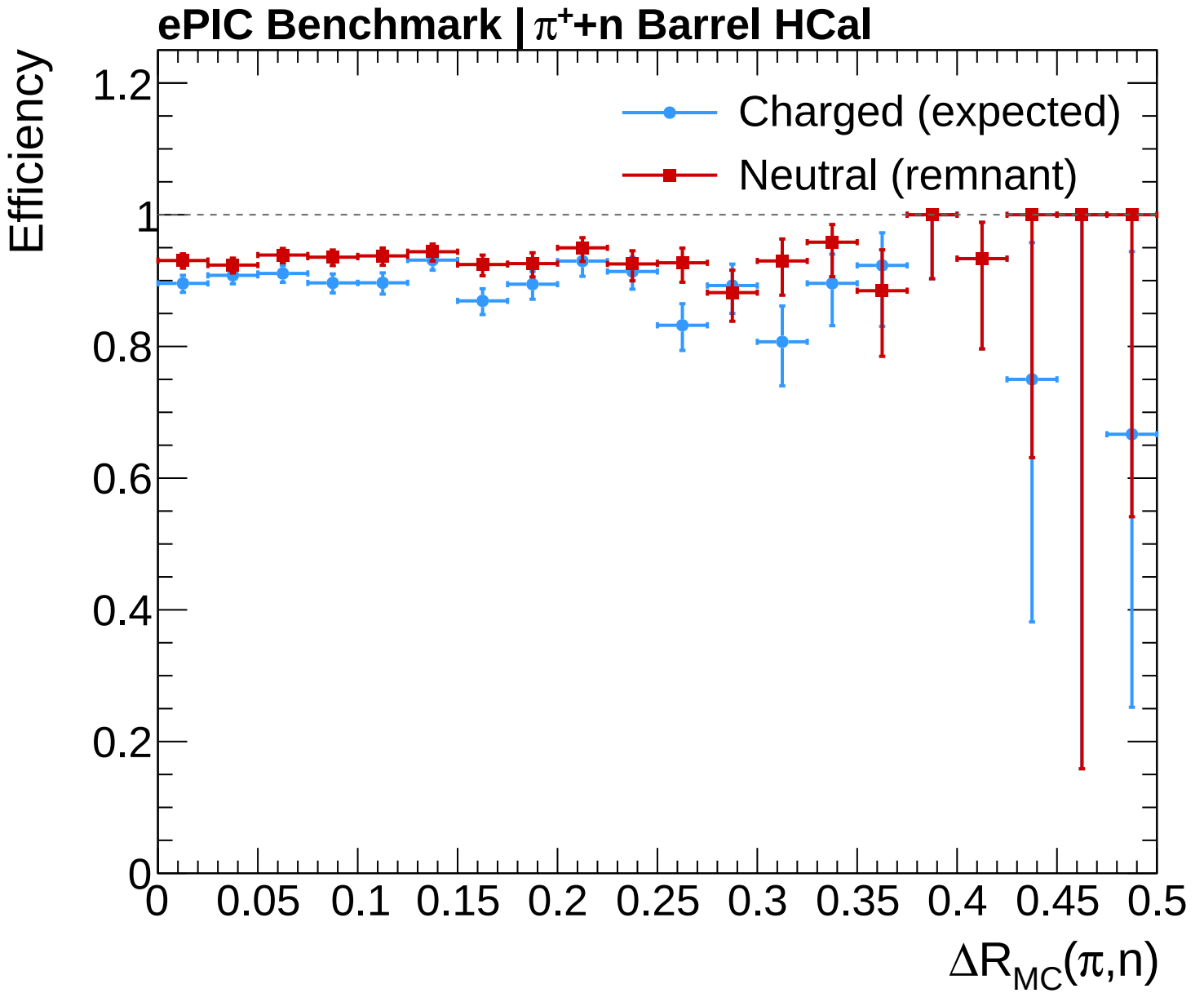

ePIC Benchmark | π⁺+n Barrel HCal
Charged: E_exp/E_MC,π
Neutral: E_rem/E_MC,n
< E_reco / E_MC >
ΔR_MC(π,n)
0
0.5
1
1.5
2
2.5
3
3.5
4
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5

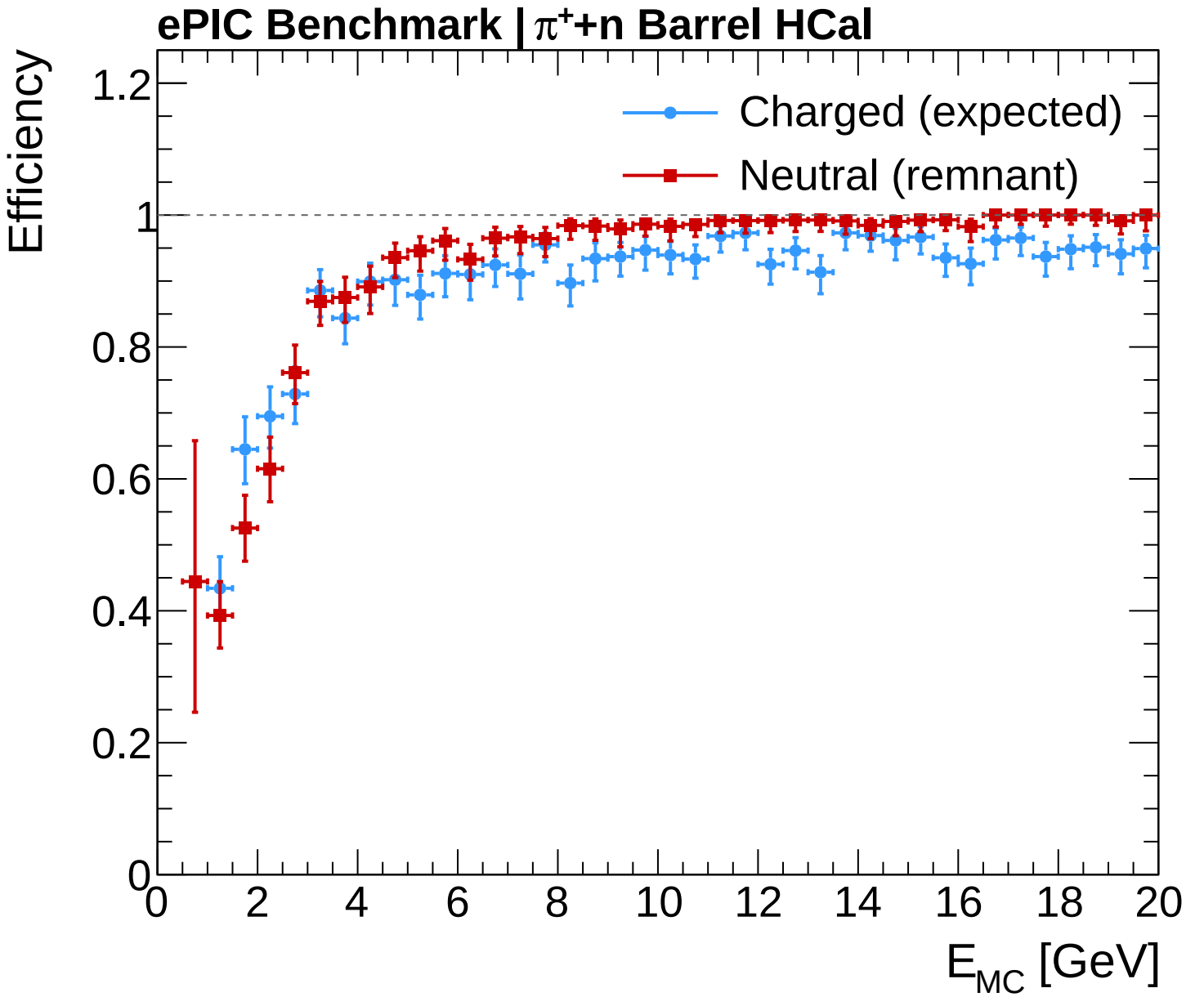

ePIC Benchmark | π⁺+n Barrel HCal
(E_exp+E_rem) / (E_MC,π+E_MC,n)
< E_reco,tot / E_MC,tot >
ΔR_MC(π,n)
0
0.5
1
1.5
2
2.5
3
3.5
4
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5

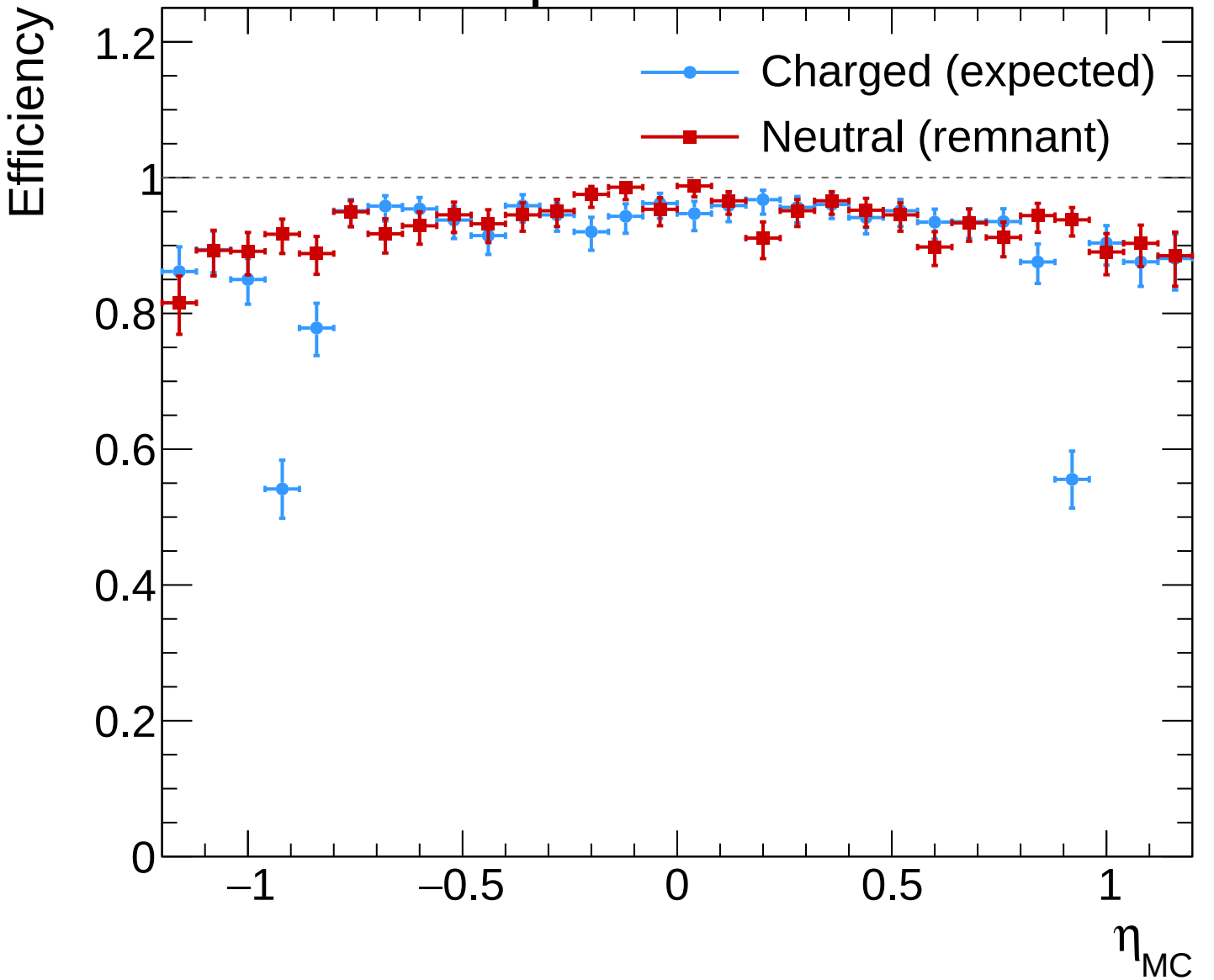

ePIC Benchmark | π⁺+n Barrel HCal
< E_exp / p_Tr >
E_exp / p_Tr
4
3.5
3
2.5
2
1.5
1
0.5
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(π,n)


ePIC Benchmark | π⁺+n Barrel HCal
Charged: E_exp/E_MC,π
Neutral: E_rem/E_MC,n
< E_reco / E_MC >
η_MC
4
3.5
3
2.5
2
1.5
1
0.5
0
–1
–0.5
0
0.5
1


ePIC Benchmark | π⁺+n Barrel HCal
Charged: E_exp/E_MC,π
Neutral: E_rem/E_MC,n
< E_reco / E_MC >
E_MC [GeV]
0
0.5
1
1.5
2
2.5
3
3.5
4
0
2
4
6
8
10
12
14
16
18
20


ePIC Benchmark | π⁺+n Barrel HCal
Efficiency
Charged (expected)
Neutral (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(π,n)


ePIC Benchmark | π⁺+n Barrel HCal
Efficiency
Charged (expected)
Neutral (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
0
2
4
6
8
10
12
14
16
18
20
E_MC [GeV]


ePIC Benchmark | π⁺+n Barrel HCal
Efficiency
Charged (expected)
Neutral (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
–1
–0.5
0
0.5
1
η_MC